Bintang Bennaya Hasiholan Hutabarat

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EDUCATION

Indonesian Institute of Technology South Tangerang, Indonesia
Bachelor of Informatics Engineering (GPA: 3.10/4.0) *2022 - Undergraduate*

SMK Letris Indonesia 2 South Tangerang, Indonesia
High School Diploma in Network Engineering (Average Grade: 88.00) *2019-2022*

WORK EXPERIENCE

CV. Citra Fire Alarm Cisauk, Indonesia
System Integration Engineer *Nov 2025 - Present*

- Engineered cross-platform Human-Machine Interface (HMI) applications for industrial panel controllers using Flutter, parsing low-level raw telemetry data received via RS-485 serial communication protocols.
- Developed real-time serial data parsing algorithms and custom cryptographic decryption modules to ingest, validate, and process encrypted hardware signals from Master Control Fire Alarm (MCFA) panels.
- Architected a centralized multi-tenant emergency alerting infrastructure utilizing Next.js, NestJS, and Tauri (Rust), automating instant incident notifications via WebSockets, WhatsApp API, and Telegram API integration.

CV. Miftah Digital Solusi Jakarta, Indonesia
Fullstack Developer *June 2025 - November 2025*

- Architected and engineered an end-to-end Garment Inventory & Supply Chain Tracking System using Laravel, tracking raw material lifecycle from warehouse entry to retail store distribution.
- Designed and implemented a real-time barcode scanning and generation module to track item movements across production stages, reducing manual data entry errors and inventory discrepancies.
- Built responsive administration dashboards, relational MySQL database schemas, and highly optimized database queries to deliver real-time stock auditing and supply chain data analytics.
- Developed secure RESTful API endpoints and integrated role-based access control (RBAC) to ensure secure operational workflows for warehouse operators and retail store managers.

PT Menara Indonesia - Kampus Gratis Jakarta, Indonesia
Cybersecurity - Internship *February 2025 - June 2025*

- Provisioned, hardened, and maintained production Linux Virtual Private Servers (VPS), configuring Nginx reverse proxies, SSL/TLS certificates, and UFW firewall security policies.
- Established comprehensive real-time infrastructure monitoring pipelines by integrating Prometheus and Grafana dashboards to track CPU, memory, network I/O, and server performance metrics.
- Executed automated performance and stress testing scenarios using K6, simulating heavy user concurrency to identify system bottlenecks and optimize API response times under high-load conditions.
- Architected an AI-powered workspace and task management platform (Notion-style) using Laravel, integrating LLM APIs for automated work logging, task summarization, and deployment.
- Enhanced the official company web portal by engineering a dynamic Content Management System (CMS), enabling seamless content updates, SEO management, and media administration.

PT Kensington Teknologi Jakarta, Indonesia

Backend Developer - Internship *September 2024 - February 2025*

- Architected scalable RESTful API microservices using Node.js and Express.js, strictly adhering to Software Development Life Cycle (SDLC) best practices and Agile methodologies.
- Engineered core backend infrastructure for an enterprise travel ticketing platform, handling booking management, dynamic pricing calculations, and transaction state workflows.
- Collaborated closely with cross-functional frontend teams to integrate APIs, streamline data contracts via Postman, and implement secure JWT authentication mechanisms.
- Managed end-to-end server deployment pipelines, environment configuration, database migration strategies, and error-logging mechanisms for high-availability staging environments.

PROJECTS

Seedrym.io | https://seedrym.io | Next.js, Rust (Axum), Solana, Supabase

- Architected a Web3 Real-World Asset (RWA) tokenization platform as a Fullstack Engineer, leveraging Next.js for reactive frontend interfaces and Rust (Axum) for high-performance backend microservices.
- Implemented secure cryptographic wallet authentication mechanisms and integrated low-latency API routes to process decentralized asset transaction pipelines.
- Managed smart contract state synchronization and backend validation logic to ensure secure real-time token transactions.

FireCo - Fire Monitoring System | Industrial IoT | Next.js, Rust (Tauri), RS-485, PLC

- Engineered a real-time industrial fire alarm monitoring desktop and mobile system deployed across vital industrial facilities, including PT PLN UP2B DKI Jakarta & Banten, Baker Hughes, and PT Surya Rengo Container.
- Integrated low-level RS-485 serial communication protocols to interface directly with hardware Master Control Fire Alarm (MCFA) panels for raw telemetry ingestion.
- Developed custom telemetry parsing algorithms and real-time decryption logic to display live alarm status metrics and system health diagnostics.

DDS Central-App | Security Token & Infrastructure | Next.js, Nest.js, MySQL

- Architected a central management server that dynamically generates cryptographic SHA-256 validation tokens to authenticate and secure remote FireCo client installations.
- Implemented automated log aggregation and remote monitoring pipelines to pull application logs from distributed sites for zero-downtime maintenance and rapid troubleshooting.
- Engineered centralized remote controls to trigger multi-channel emergency alert dispatches via WhatsApp API and Telegram API integrations.

SKILL

- **Programming Languages**: TypeScript, Javascript, Rust, SQL (MySQL, PostgreSQL)
- **Framework Libraries**: Next.js, NestJS, Node.js, TailwindCSS, Laravel, Tauri, Axum
- **Database & Infrastructure**: MongoDB, MySQL, PostgreSQL, Firebase, Prisma
- **Developer Tools**: Git, Websockets, MCP Servers, Postman, K6
- Professional Competencies: AI-Assisted Development, Public Speaking, Engineering, Empathy, Collaborative, Problem Solving, Technical Mentoring, Project Scoping